Progress since the October 2 advisor meeting

Survival supervision, feature sensitivity and multimodal evidence

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2026-10-15

What the work has established

The approved October 14 write-up remains the project's benchmark. Its single-block model has mean external concordance **0.627** across five cohorts, with $\alpha_F = 0$: survival informs the coefficients but not the loading updates. The matched expression–methylation framework has ICGC primary-PDAC concordance **0.685**, without demonstrated improvement over expression alone. Neither result has been replaced.

Since that write-up, the experiments have produced positive evidence that survival-informed fitting can recover a low-variance prognostic program in simulation, a clean simulation comparator panel, and reproducible diagnoses of real-data instability. They have **not established a final supervised method** satisfying convergence, rank selection and external performance requirements. Several central feature-sensitivity requests are still open.

October 2 request	Evidence available	Status
Unified methods/results/biology write-up	Approved interim PDF and reproducible frozen results	Complete as an interim account
Survival shapes loadings F	Implemented experimentally; replicated recovery evidence	Partial; adoption requirements fail
More features at fixed K	Limited expression experiment, 662–3,600 shared genes	Full-range curves and agreement remain open
Probe-wise methylation variability	Full 305,492-probe observed-value inventory and SD histograms	Complete descriptively; filter choice remains open
Parsimonious retained programs	Protected/full-state pruning evaluated and rejected	Rank selection unresolved
Multimodal value	Methylation-dominant and shared molecular structure	Predictive benefit and validated added biology unproven

What is concretely solved. Dead point-mass programs no longer receive survival protection in the opt-in pruning rule. Full posterior information can be retained through opt-in pruning refits. Comparator rank-zero/rank-one cases and individual errors remain in denominators. The failed historical framework CV fold is exactly reproducible. A curvature experiment removes constant-risk predictions in its five candidate CV folds. These are specific software corrections and diagnoses, not proof of the complete scientific goal.

Current decision. No replacement is proposed. Preserve the benchmark, report failures, and now prioritize the advisor's fixed-K feature study alongside bounded method remedies. Andrew approved that revised sequence.

Does survival now influence F?

In experimental code, yes. The canonical single-modality update combines reconstruction and survival precision/signal contributions, $A_F = A_{\text{gen}} + w_S A_{\text{surv}}$ and $B_F = B_{\text{gen}} + w_S B_{\text{surv}}$, with $w_S = 1$ in the evaluated unweighted formulation. It also fixes projection scale and can use survival-informed initialization. The benchmark's $\alpha_F = 0$ update omits those survival contributions. It remains close to factorization followed by survival regression in this important sense, although its stopping rule means its loadings need not equal a separately converged unsupervised fit.

Recovery evidence and its limits

The clean comparator rerun covers 40 simulated datasets and 200 method results, with no fitting errors. Its supervised arm recovers the planted program in 17/20 low-noise and 19/20 high-noise datasets. The four comparator arms do not recover the planted survival-active program under the specified joint criterion; EBMF alone has no fitted survival risk by design. Two supervised fits reach their iteration cap, and one high-noise prediction is constant. This is controlled simulation evidence, not an external-cohort result.

The latest, separately approved profiled-curvature experiment changes only an opt-in flag from the framework controls. Its simulations give:

Model	Noise	Recovery /20	False active	Reference false active	Converged /20
Single	.25	18	1.35	1.15	19
Single	1	20	.40	.35	20
Multimodal	.25	16	.80	.75	20
Multimodal	1	19	.40	.65	20

All recovery and false-positive criteria pass; Single convergence does not. The fixed beds are retained. These results do not establish validity for all feature counts, rank selection or the final Bayesian objective.

External performance has not established non-inferiority

The earlier selected supervised Single candidate, evaluated before Round 2, was compared with the frozen D4 reference on five cohorts (616 patients).

Model	Mean external C	Delta versus D4 (95% interval)
Frozen D4, $\alpha_F = 0$	.626699	Reference
Earlier supervised candidate	.620725	$-.005974$ $[-.031708, .020662]$
Earlier $\alpha_F = 0$, point-Laplace L/F arm	.635304	.008605 $[-.011261, .030378]$

The supervised arm fails the required lower limit > -0.02. The earlier $\alpha_F = 0$ arm passes that requirement but does not supervise F.

The latter arm changes loading priors as well as retaining $\alpha_F = 0$; it is not the frozen D4 fit. No model is selected from these external scores. No new Round 2 external cohort has been scored. Simulation recovery alone therefore does not close the supervision requirement.

Multimodal stability and the meaning of seven programs

The selected supervised multimodal fit collapsed to zero risk on TCGA. Training-only tracing identified shared coefficient-prior absorption. Replacing it by a fixed prior did not produce a converged, credibly prognostic real-data fit. An unchanged historical-framework CV fold also lost all coefficient means and second moments; the exact trace reproduces all 13 saved-state/history checks, locating the persistent loss at outer iteration 167, inner sweep 1.

The approved working-intercept profile removes a penalty on the arbitrary constant-risk direction. All five candidate CV risks are nonconstant, but all five reach 500 iterations, and credible-program counts are 0,1,0,0,0. Its own five-fold mean C is .563678. The original reference has one undefined score, so the prescribed five-finite-pair comparison remains ineligible. Single candidate CV C=.624776 versus .623985 passes its paired one-SE rule.

Source/model, full TCGA	Training C	Credible programs	Final K	Converged
Frozen joint framework	.637768	1	7	No
Frozen expression-only framework	.643559	1	7	No
Frozen matching Single $\alpha_F = 0$ comparison	.593717	—	—	—
Round 2 profiled joint framework	.639172	1	7	No, 500 cap
Round 2 profiled Single framework	.651632	3	7	No, 300 cap

These are training quantities, not the external benchmark values .685/.627. The profile panel completes 92 new fits plus 10 cached references without execution errors or warnings. Eight new fits cap. It is not adopted.

Seven may be appropriate; the roles must justify the count

A joint fit can retain expression-dominant, methylation-dominant and prognostic programs, with overlapping roles. Fewer programs are not inherently better. The frozen joint fit already provides this descriptive role evidence:

Program	Provisional interpretation	Expression share	Methylation share	Survival-active
1	Classical/gastric	23.5%	4.8%	No
2	Immune	5.1%	20.9%	No
3	Mixed, unlabelled	31.5%	0.3%	No
4	Basal-like versus classical	6.2%	4.5%	Yes
5	Normal versus activated stroma	4.8%	1.2%	No
6	Exocrine/acinar	28.9%	9.3%	No
7	Methylation-only	0%	58.9%	No

All rows are from the frozen benchmark, not inferred labels for the newer fit. Shares normalize fitted rank-one squared norms within a modality; they are not additive variance explained or survival contributions. Labels are provisional; CpG annotation and enrichment were not performed.

The Single write-up's four retained programs classify one K=7 fit; three other fitted programs are inactive under its operational thresholds. No deletion/refitting produced those four. Thus "7 versus 4" compares classification/model settings as well as modalities. A matched role and redundancy audit is required before interpreting it as excess joint rank.

How much of the feature-count request is complete?

A limited earlier Single experiment varied the screening count per cohort, then intersected the selected features. Requested count and fitted dimension are different quantities:

Requested/cohort	Shared genes	Mean training CV C	CV SE	Capped folds	Frozen D4 external C
1,000	662	.650200	.017200	1/5	—
2,000	1,365	.648933	.022598	0/5	—
3,000	2,064	.651199	.023631	1/5	—
5,000	3,600	.655413	.027386	2/5	—
Frozen D4 reference	2,064	—	—	—	.626699

This earlier curve uses the selected experimental Single configuration: canonicalization, $w_S = 1$, supervised initialization and a shared point-Laplace coefficient prior. It is not the frozen $\alpha_F = 0$ D4 configuration. All four full-training fits converge. The observed differences are small relative to their uncertainty and do not justify a new feature count. Screening/scaling preceded folds; these are conditional CV results, not a fully nested raw-data pipeline. The runner exited nonzero after saving its tables because its source was edited during execution. Saved fits/counts and summary arithmetic were independently verified; a clean-process rerun remains optional unfinished work. This limitation is not hidden.

This experiment does **not** complete the advisor request: it stops well short of the maximum feature universe, omits CpG-count expansion and does not provide the requested factor-agreement curve.

Approved next experiment: feature count varies; K does not

Start with one modality, fixed **K=7 and deletion off**. Freeze the model, patients, ranking, folds and convergence rule, and increase nested feature sets through the full eligible universe. Report the actual number after intersection, not an assumed 20k/300k maximum. First use the benchmark/fallback as labelled sensitivity analysis; do not wait indefinitely for a supervised replacement or describe this as solving supervision.

For each set report C and held-out partial log-likelihood, convergence/caps, runtime/memory, loading shrinkage and program roles. Match programs using training activities, then correlate loadings on common features and compare projection/risk agreement. Preserve ambiguous matches and undefined correlations. Warm-start larger sets as specified; cold-fit the largest set for comparison. Do not change K, weights and feature counts together.

Next do methylation alone, then matched joint fits varying one modality at a time. External outcomes cannot select feature counts. Large CpG fits may require prepared Longleaf arrays, which will not be submitted automatically.

Full-probe methylation variability: completed

The descriptive run covers **305,492 available TCGA probes on the same 144 training patients**. It uses observed raw beta values before imputation. All probe SDs are defined; 29,414 individual values are missing and remain missing in these calculations. No survival outcome, model fit or new cutoff is used. The screened reference IDs and patient order match the frozen fit.

Descriptive set	CpGs	Overlap with frozen screened 10k
All available TCGA probes	305,492	10,000
Mean beta in [.1,.9]	182,996	9,990
Neither extreme fraction reaches 95%	221,222	10,000
Both filters	182,996	9,990
Frozen screened reference IDs	10,000	10,000

The proportion rule retains 38,226 probes excluded by the mean rule. The mean rule would also exclude 10 currently screened probes; **neither rule is adopted**. Median beta SD is **.0600** over all probes, versus **.1793** in the frozen screened set using raw observations (.1792 in the existing imputed cache). This confirms that the top-variance screen substantially enriches variability; it does not show which feature universe explains survival better.

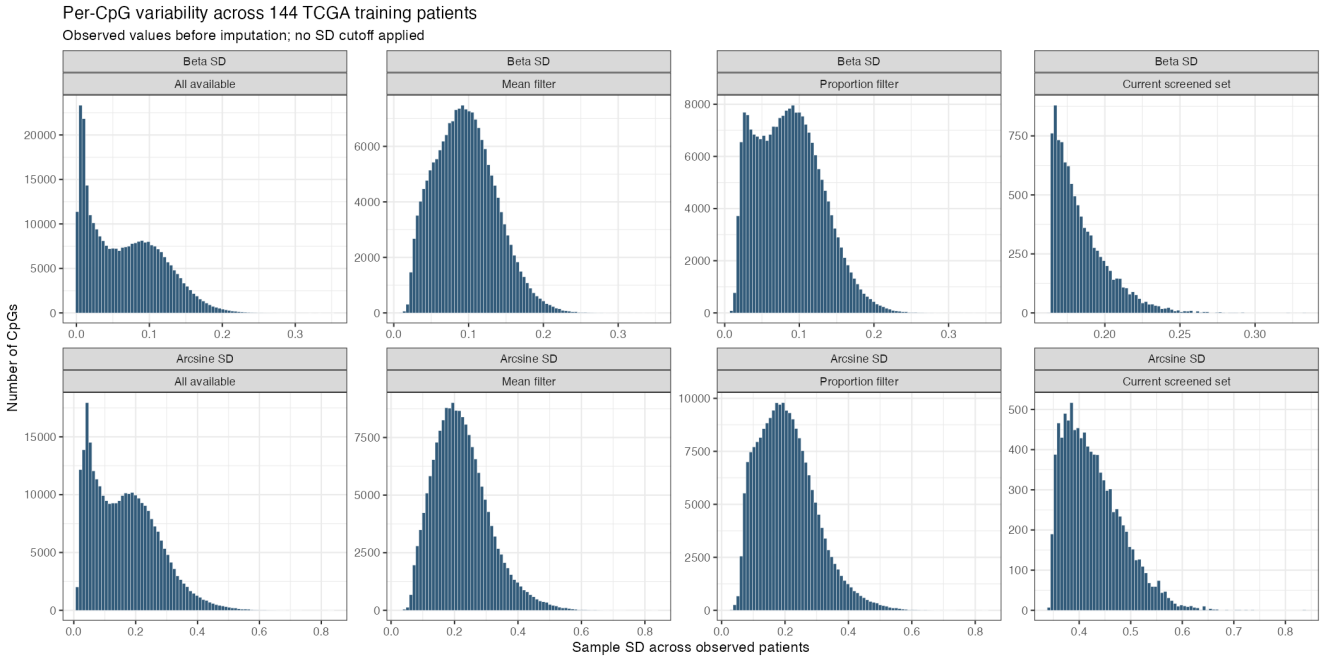


Figure 1: Per-probe sample SD across observed TCGA training patients, beta and arcsine scales. Each panel has its own axes; counts differ by descriptive group. Undefined observations are retained in the inventory, and no probe has undefined SD here. Filters are comparisons, not adopted preprocessing.

The companion mean-versus-SD density plot and safe aggregate tables are in the CpG output directory. Cross-reactive/SNP probe-filter provenance remains unverified; no new QC or imputation was applied. This completes the requested descriptive inventory and SD histogram, while the feature-count/performance experiment remains open.

Multimodal value and manuscript integration

Two acceptable multimodal contributions

Prediction: compare joint, expression-only and methylation-only fits on the same patients, folds and expression feature counts. Report paired uncertainty. Matching expression-only C is not predictive improvement, and training C alone is not external performance.

Additional biology: assess stable molecular structure absent from expression-only fitting. The methylation-only and immune components already provide descriptive candidates. Neither is survival-active in the benchmark; 58.9% methylation share is not evidence of 58.9% survival explanation. Annotation, enrichment, reproducibility across feature sets and checks of composition/technical explanations are needed for a stronger biological claim.

What enters the larger write-up

Revision dependency	Evidence available for review	Still required
R1–R3: updates, scale and objective	Implemented supervision, curvature diagnosis, prior/KL caveats	Stable defensible final formulation
R4: rank/classification	Failed pruning panels; distinction between seven fitted/four classified	Valid rank choice and matched program-role audit
R5–R7: evaluation/results	Recovery/comparator tables, external failure, limited feature CV	Full-range sensitivity and final allowed external report
R8: biology	Frozen program roles, provisional labels and limitations	Final-program CpG annotation and confounding checks
R9: implications/continuing work	Clear progress and unresolved requirements	Conclusions supported by the final reviewed evidence

The initial PDF remains unchanged. This report supplies an auditable progress account and revision evidence; it does not replace it. Further findings can be incorporated after review without presenting unresolved goals as achieved.

Evidence and provenance

All numbers above are saved results; this report runs no model or new external scoring. The benchmark tag is `benchmark-interim-writeup-2026-10-14`; all 10 snapshot hashes match. The saved model checkpoint before this report is `f0858ec`; the CpG design is committed at `dc69f6d`, with 642 passing software tests. Full per-task commits, failures and tables are indexed in the Round 2 handoff and attempt history. The approved priority plan links these deliverables to the October 2 meeting and write-up revision markers. Software tests validate specified behavior; their count is not proof that all statistical goals have been met.